

$$\textcircled{1} \quad O_{sf} = 10 \text{ mg/L}$$

$$f(T) = \ln(O_{sf}) - \ln(10) = 0$$

$$T_a = T_f = 273.15$$

$$\textcircled{a} \quad T = 0^\circ\text{C} \quad (T_a = 273.15 \text{ K})$$

$$\ln(O_{sf}) \approx 2.619 \Rightarrow f(0) = 2.619 - 2.302 = 0.317$$

$$\textcircled{a} \quad T = 40^\circ\text{C}, \quad T_a = 313.15 \text{ K}$$

$$\ln(O_{sf}) = 1.862 \Rightarrow f(40) = 1.862 - 2.302 = -0.440$$

$$\underline{\text{It 1:-}} \quad T_1 = 0 + 40 = 20^\circ\text{C} \quad (T_a = 293.15 \text{ K})$$

$$f(20) \approx \ln(9.09) - \ln(10) = -0.095$$

$$f(0) > 0, \quad f(20) < 0, \quad [0, 20] \rightarrow \text{new interval}$$

$$\underline{\text{It 2:-}} \quad T_2 = 0 + 20 = 10^\circ\text{C} \quad (T_a = 283.15 \text{ K})$$

$$f(10) = \ln(11.4) - \ln(10) = 0.120$$

$f(10) > 0, f(20) < 0$, $[10, 20] \rightarrow \text{new interval}$

Iter	a	b	c	f(c)
1	0	10.000	20.000	-0.0954 $[0, 20]$
2	0	20.000	10.000	0.1201 $[10, 20]$
3	10.000	20.000	15.000	0.0076 $[15, 20]$
4	15.000	20.000	17.500	-0.0450 $[15, 17.5]$
5	15.000	17.500	16.250	-0.0190 $[15, 16.25]$
6	15.000	16.250	15.625	-0.0058 $[15, 15.625]$

As we continue,

with my python code , the bisection method

converges to a final freshwater $T = 15.3882^\circ\text{C}$.

② $Q = 5 \text{ m}^3/\text{s}$ $B = 20 \text{ m}$, $S = 0.0002$

$\eta = 0.03$.

$$f(y) = \frac{\sqrt{0.0002} (20y)^{5/3}}{0.03 (20+2y)^{2/3}} - 5 = 0$$

$$f'(y) = \frac{\sqrt{0.0002} (20)^{5/3} (y^{2/3}) (20+y)^{-5/3}}{0.03} \left[\frac{5}{3} (20+2y) \right]$$

It is:- $y_0 = 1.0 \text{ m}$ (Initial guess).

$f(1.0)$:

$$f(1.0) = \frac{0.01414 (20)^{5/3}}{0.03 (22)^{2/3}} - 5 = 8.847 - 5 = 3.847$$

$$f'(1.0) = 0.4714 (143.36) (1.0) (22)^{5/3} [33.33 + 2] = 14.165$$

$$y_1 = y_0 - \frac{f(y_0)}{f'(y_0)} = 1.0 - \frac{3.847}{14.165} = 0.728 \text{ m}$$

St 2 : $y_1 = 0.728 \text{ m}$

$$f(0.728) = \frac{0.01414 (14.56)^{5/3}}{0.03 (21.456)^{2/3}} - 5 = 5.293 - 5 = 0.293$$

$$f'(0.728) \approx 1.79$$

$$y_2 = y_1 - \frac{f(y_1)}{f'(y_1)} = 0.728 - \frac{0.293}{1.79} = 0.703 \text{ m}$$

$\therefore y \approx 0.703 \text{ m}$ (and continued for another iteration).

The newton Raphson method converges rapidly.

The final flow depth for discharge of $5 \text{ m}^3/\text{s}$

is $y = 0.7034 \text{ m}$.